

Lab: Getting Started with draw.io for Systems Modeling

Estimated time: 20 minutes

Learning objectives

After completing this lab, you will be able to:

- Open and use draw.io
- Create a context diagram
- Build a level 0 data flow diagram (DFD)
- Create a use case diagram

Introduction

draw.io (also known as diagrams.net) is a free, web-based diagramming tool used to create visual models such as flowcharts, system diagrams, and data flow diagrams. With drag-and-drop functionality and a wide variety of shapes and connectors, draw.io is ideal for software design, process mapping, and systems architecture.

In this lab, you'll learn to model a simple Library Management System (LMS) by creating three common systems architecture diagrams: a Context Diagram, a DFD (Level 0), and a Use Case Diagram. No sign-up is required.

Exercise 1: Launch draw.io

Step 1: Launch your browser and go to <https://app.diagrams.net/>.

Step 2: In the pop-up that appears, select **Create New Diagram**.



app.diagrams.net



Microsoft Azure



Adobe Acrobat



File

Edit

View

Arrange

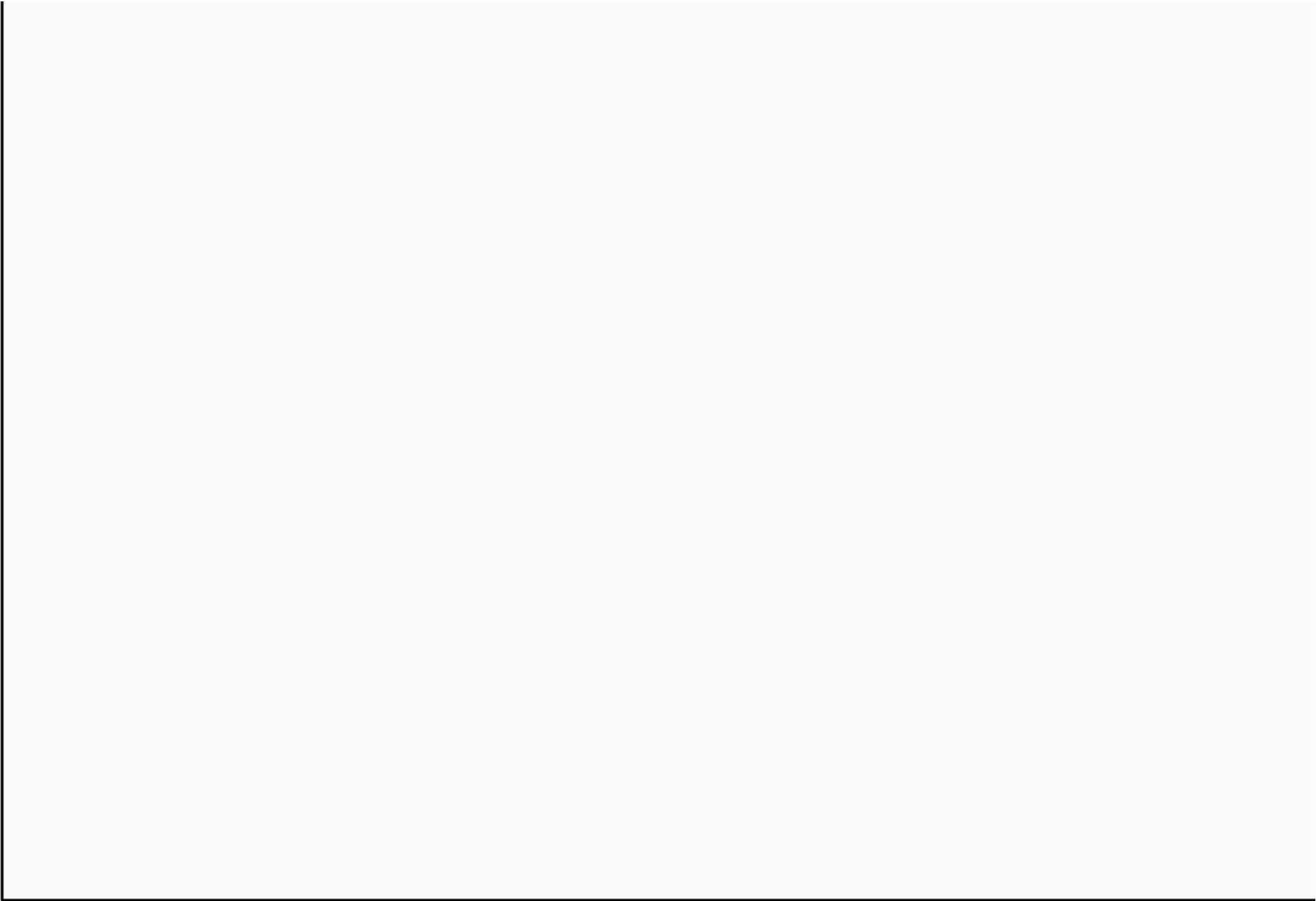
Extras

Help



100% ▾





***Note:** If you do not see this pop-up, Steps 3 to 6 do not apply. You can proceed directly to working in the main diagram interface (Step 7).*

Step 3: In the filename field on top, provide a suitable name for your diagram.

Step 4: Then, from the drop-down list at the top-right of the pop-up, select the format in which you want to save your diagram. For example, **Editable Bitmap Image (.png)**.

Step 5: Select **Blank Diagram**. Then, at the bottom-right of the pop-up, select **Create**.



app.diagrams.net



Microsoft Azure



Adobe Acrobat



File

Edit

View

Arrange

Extras

Help



100% ▾



Filena

Search

Basic (12)

Smart Templ

Business (15)

Charts (6)

▶ Cloud (41)

Engineering
Flowcharts (9)
Maps (5)
Network (13)
Other (12)
Software (12)
Tables (4)
UML (8)
Venn (8)

Step 6: When the save window appears, choose your preferred location (for example, Desktop), confirm the filename (Lab.drawio), and then select **Save** to store your diagram.



Warning: this site can see edits you make



> Sharon - Fl... > Desktop >



Organize ▾

New folder



Home



Gallery



Sharon - Flexible



Desktop



Downloads



Documents



Name

File name: Lab.drawio

Save as type: Editable Bitmap Image

^ Hide Folders

Network (13)

Other (12)

Software (12)

Tables (4)

UML (8)

Venn (8)



Step 7: The draw.io workspace is now open and ready. You can start modeling your diagram by selecting the desired shapes from the left pane and customizing them as needed on the canvas (in the center pane).



Lab.drawio.png

File Edit View Arrange Extras Help All changes saved



100% ▾



Search Shapes

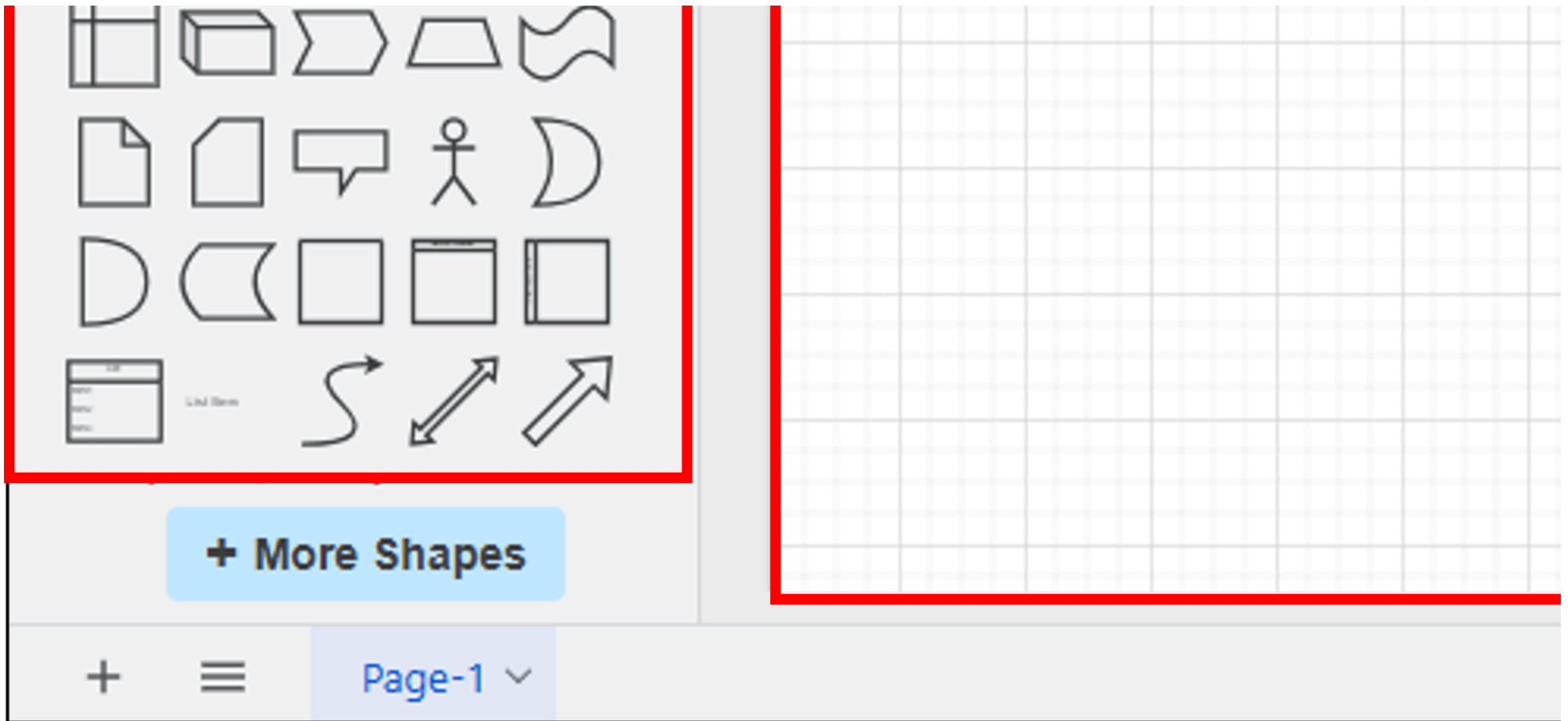


▼ Scratchpad ⓘ + ✎ ✕

Drag elements here

▼ General





Exercise 2: Create a context diagram

A context diagram shows a high-level view of your system and how it interacts with external entities.

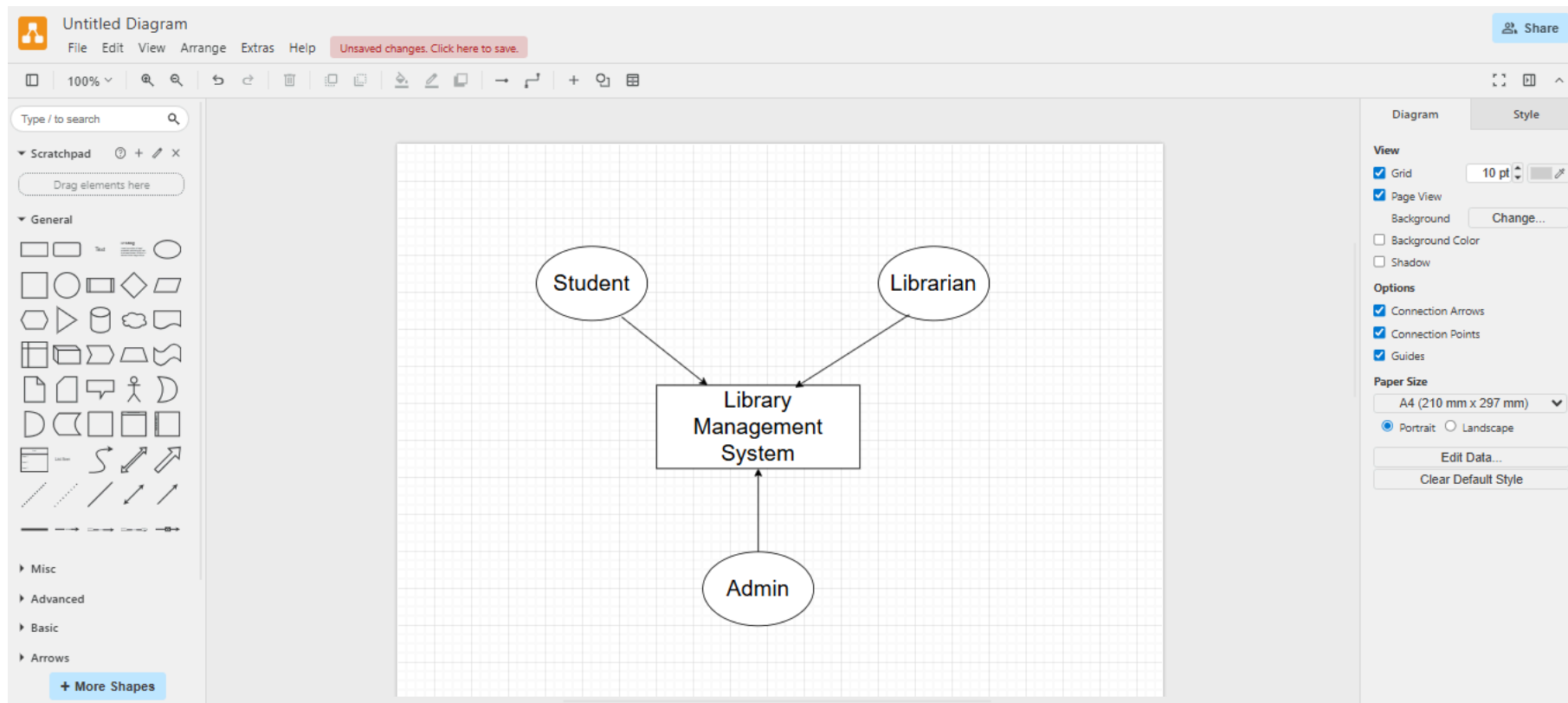
Step 1: In the center of the page, draw a rectangle and label it "**Library Management System**".

Step 2: Around the Library Management System system, add three external entities (actors) using ovals:

- Student
- Librarian
- Admin

Step 3: Connect each entity to the **Library Management System** using arrows to indicate data flow.

Note: Your diagram should look similar to a hub with the **Library Management System** in the middle and the three external entities around it.



Exercise 3: Create a level 0 DFD

A Level 0 DFD breaks the system into major processes and shows how data flows between entities, processes, and data stores.

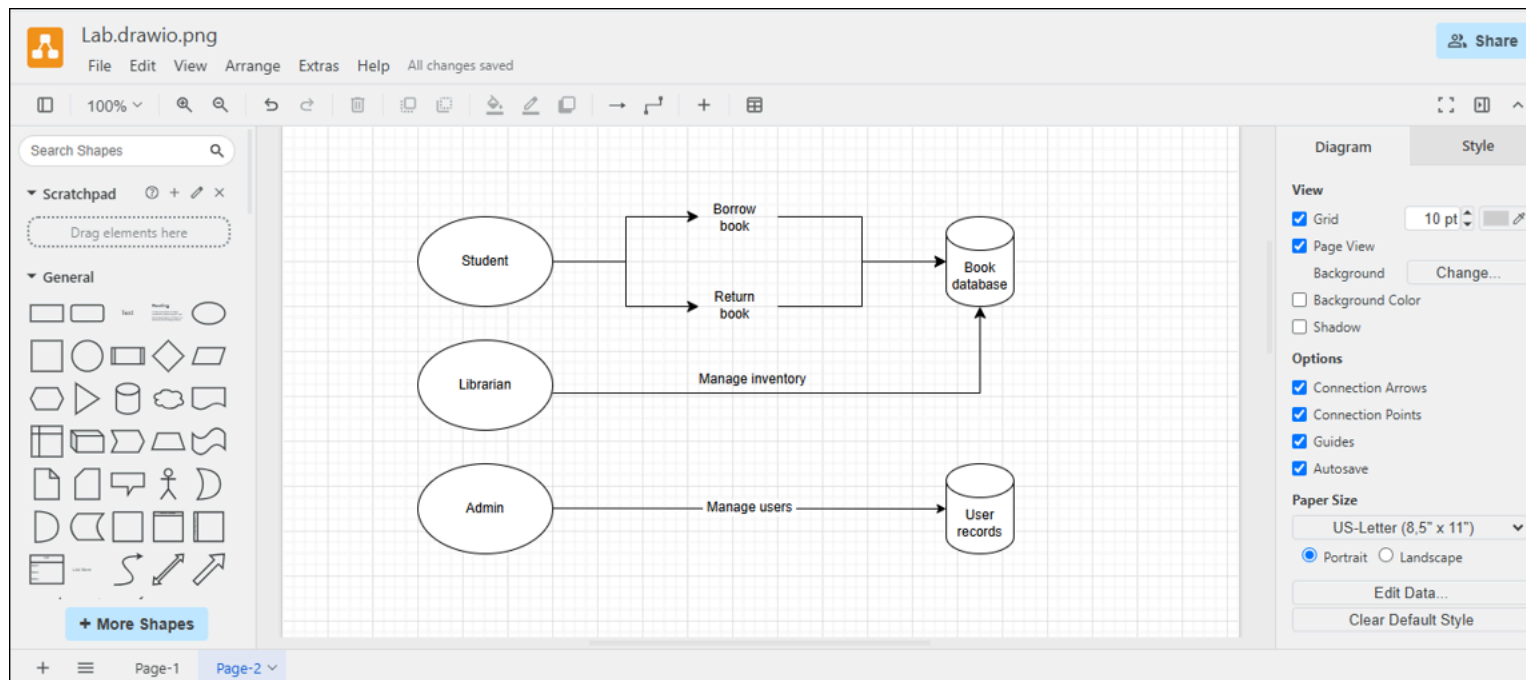
Step 1: Add a new page by selecting the + icon at the bottom-left corner.

Step 2: In Page-2, add the external entities (actors):

- From the left shapes panel, under **General**, select the Ellipse shape thrice.
- Double-click the ellipses on the canvas to label them:
 - Student
 - Librarian
 - Admin

Step 3: Add processes, databases, and connectors:

- From the left shapes panel, under **Misc**, select the Horizontal Elbow arrow and add two such arrows from the Student entity, as shown in the diagram below. Label the arrow heads using the Text box from the left shapes panel, under **General**:
 - Borrow book
 - Return book
- Then, from the left shapes panel, under **General**, select a Cylinder and move it to be in the same line as **Student**, as shown in the diagram below. Label the cylinder "**Book database**".
- Use two Horizontal Elbow arrows from "**Borrow book**" and "**Return book**" to reach the **Book database** cylinder as shown.
- Next, from the **Librarian** entity, use the Horizontal Elbow arrow to connect the **Librarian** ellipse and the **Book database** cylinder. Add the text "**Manage inventory**" as shown.
- Then, from the left shapes panel, under **General**, select a Cylinder and move it to be in the same line as **Admin**, as shown in the diagram below. Label the cylinder "**User records**".
- Finally, from the left shapes panel, under **General**, select a Connector with Label and connect the **Admin** entity to **User records**.
- Double-click the label in the center and enter the task "**Manage users**" as shown.



Exercise 4: Create a use case diagram

Use case diagrams show which actors interact with which system functions.

Step 1: In a new page, in a similar manner to the DFD diagram, draw shapes for each external entity (actor):

- Student
- Librarian

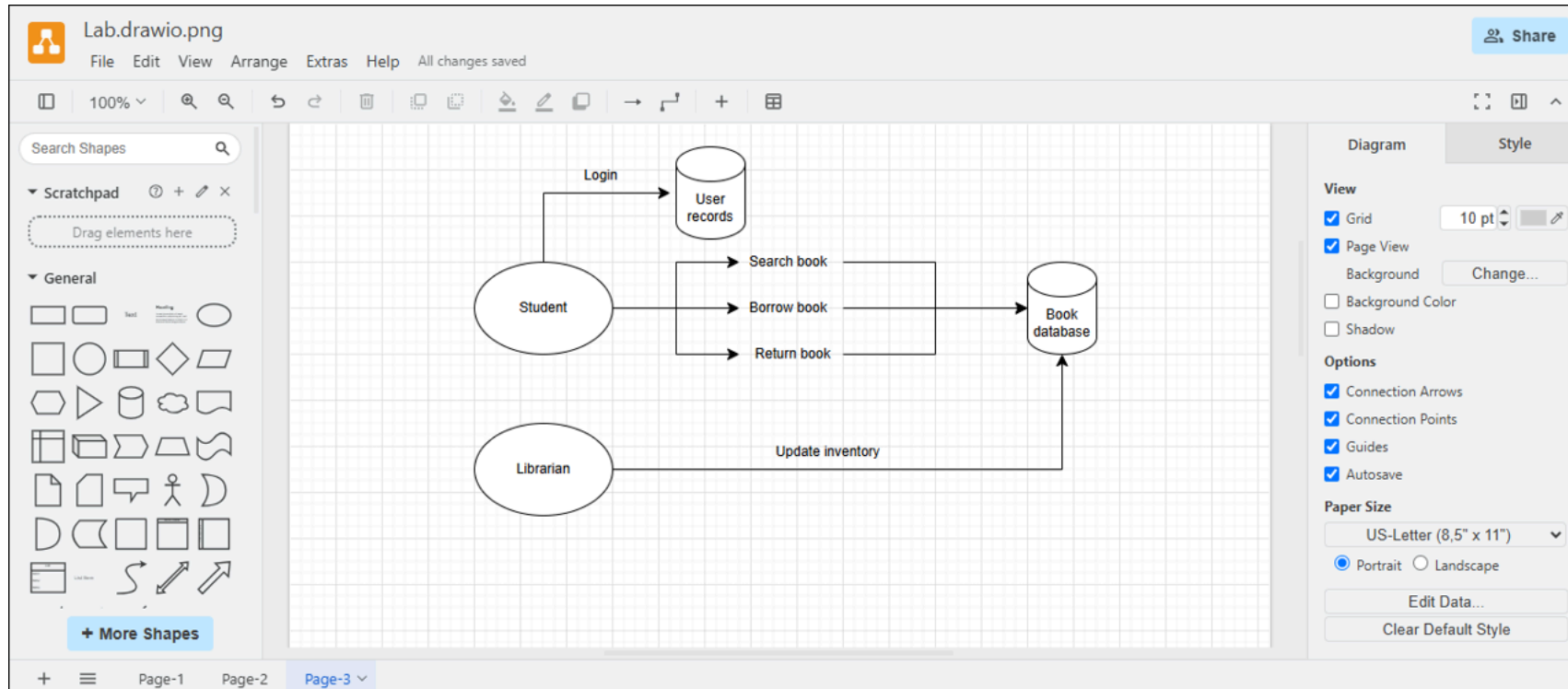
Step 2: Add textboxes and cylinders for each system action:

- Login
- Search book

- Borrow book
- Return book
- Update inventory
- User records
- Book database

Step 3: Connect each actor to the relevant use cases as shown in the diagram below:

- Student → Login → User records
- Student → Search book, Borrow book, Return book → Book database
- Librarian → Update inventory → Book database



Summary

You've now created three key systems diagrams using draw.io:

- A context diagram for high-level interaction
- A level 0 DFD showing processes and data flow
- A use case diagram detailing user actions

Author(s)

Parikshit Jain